Chi (Faye) Zhang

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Education

The City University of New York (CUNY), Graduate Center — New York, NY
Ph.D. student in Computer Science | GPA: 3.73/4.0 — August 2024 – Present

Shandong University — Jinan, China
B.E. in Software Engineering | GPA:3.70/4.0 — September 2019 – June 2023

Research Interests

Security and privacy of agentic AI systems; accountability and digital forensics for AI agents; cloud audit and logging measurement; usable security; LLM-assisted education

Honors and Awards

- ACM CCS Individualized Cybersecurity Research Mentoring (iMentor) Workshop, Cohort — 2023
- Outstanding Student Award, School of Software, Shandong University — 2022
- Innovation and Entrepreneurship Scholarship, Shandong University — 2020, 2021, 2022

Selected Publications

- **Zhang, C.**, Ji, P. Toward Responsible Agentic AI Systems: Measuring the Audit Evidence Behind Agent Actions in the Cloud. Under review.
- **Zhang, C.**, Ji, P., Zhang, Z. (2026). "We Have Never Been Attacked": Unpacking the Illusion of Cybersecurity in Small Business. *Symposium on Usable Privacy and Security (SOUPS 2026) Poster*.
- Hanon, J., **Zhang, C.**, Ji, P. (2026). Privacy Tradeoffs in the Digital Era: Public Views on Data Use, Monitoring, and Accountability. *Symposium on Usable Privacy and Security (SOUPS 2026) Poster*.
- **Zhang, C.**, Liu, Y., Chen, X., Ji, P. (2026). What Keeps Agent Skills from Being Reusable? Evidence from 138K SKILL.md Files. *First Workshop on Agent Skills (AgentSkills), ACM CAIS 2026*.
- Chen, X., **Zhang, C.**, Ji, P., Liu, Y. (2026). SkillsMetric: Mapping the Detection Boundary of Static Analysis for Malicious Agent Skills. *First Workshop on Agent Skills (AgentSkills), ACM CAIS 2026*.
- Cai, J., **Zhang, C.**, Ji, P. (2025). InsightFlow: A Generative AI Approach to Streamlining Knowledge and Learning Paths. *International Conference on Web-Based Learning (ICWL 2025)*.
- Baravkar, S., **Zhang, C.**, Hassan, F., Cheng, L., Song, Z. (2024). Decoding and Answering Developers' Questions About Web Services Managed by Marketplaces. *IEEE International Conference on Software Services Engineering (SSE 2024)*, 194–203.

Research Experience

Accountability of Agentic AI Systems in the Cloud — Sep 2026 – Present
Supervisor: Prof. Ping Ji | Paper under review — New York, NY

- Conducted a large-scale measurement of audit evidence for LLM agent actions on Google Cloud and AWS, combining controlled agent deployments with scans of public agent and infrastructure-as-code repositories.
- Found that even with full logging, provider records attribute agent actions to the credential rather than the agent and cannot link an action to the model decision behind it; few public projects enable even the records that exist.
- Designed a tamper-evident agent-side record that links each action to its decision through identifiers that provider logs also carry, and validated it against provider records on both clouds.

Usable Security for Small Businesses — Oct 2024 – Present
Student Principal Investigator | Supervisor: Prof. Ping Ji | Full paper under review — New York, NY

- Designed and led a mixed-methods study of small-business cybersecurity ($N = 128$: interviews, surveys, and 102 founder intercepts), conducting all data collection, qualitative analysis, and writing.

- Found that owners often report never being attacked even though 81% of small businesses reported a breach in the previous year (ITRC 2025), and traced this **security illusion** to semantic narrowing, structural invisibility, and delegated trust.
- Derived design implications such as default-on audit signals and clear cloud responsibility boundaries.

OPAL: Open Platform for Adaptive Learning Analytics Sep 2026 – Present

PI: Prof. Ping Ji | Designer and developer New York, NY

- Designed and built **OPAL**, an **open** platform for **adaptive** learning: open to any course, because teachers bring their own slides and notes, and adaptive, because each student's practice follows their own mastery. In use at Hunter College and several secondary schools.
- For teachers: AI-drafted quizzes in which every question cites its source in the course materials, a review step so nothing reaches students without approval, and analytics showing which concepts the class struggles with.
- For students: self-paced practice that tracks mastery of each concept, points to weak spots, and never affects grades.

Reusability and Security of Agent Skills Mar 2026 – May 2026

Self-initiated | Supervisor: Prof. Ping Ji | Two workshop papers (ACM CAIS 2026) New York, NY

- Measured 138,133 public SKILL.md files from 20,556 repositories against a defect taxonomy grounded in the official specification, finding that 91.8% contain at least one defect, mostly packaging problems such as weak routing metadata (first author).
- Showed with a routing stress test over 20,000 skills that these defects hurt retrieval, and proposed a quality-assured generation workflow (spec-aware prompting, linting, repair, safety gating).
- Co-developed **SkillsMetric**, a static analysis framework for malicious skills (AUC 0.93 across 16 attack types), and identified blind spots: host-destruction attacks (0% detected) and prompt injection (42%).

LLM Agents for Cyber Defense Nov 2025 – Feb 2026

Supervisor: Prof. Ping Ji | Poster (AIX Summit 2026) New York, NY

- Built and configured multi-agent experimental environments based on CybORG CAGE Challenges, conducted prompt ablation experiments and data analysis benchmarking GPT-4o and GPT-5.2 as autonomous Blue defenders.

AeDA Platform Development Sep 2024 – Present

Supervisor: Prof. Ping Ji New York, NY

- Full-stack developer for **AeDA** (Attainable eDucation for All), an AI-powered platform for personalized learning paths; built the mobile app and web portal and architected its data storage on GCP.

Knowledge Tracing for Personalized Learning Jul 2025 – Dec 2025

Supervisor: Prof. Ping Ji | Published (ICWL 2025) New York, NY

- Curated datasets and co-wrote the manuscript for a graph-aware knowledge tracing framework that combines educator-curated knowledge graphs with Transformer models in **AeDA**.

Research Assistant, University of Michigan Aug 2023 – Dec 2023

Web Services Research Ann Arbor and Dearborn, MI

- Classified developer questions about marketplace-managed web services to improve API documentation (IEEE SSE 2024).

Research Assistant, Shandong University Nov 2020 – Jun 2023

Supervisors: Prof. Lizhen Cui, Prof. Yonghui Xu Jinan, CN

- Co-authored a survey on music intervention (IJCS 2023) and built a rhythm-matching algorithm (ICCSE '21).
- Digitized Traditional Chinese Medicine manuscripts via image preprocessing, OCR, and data annotation.

Teaching

- CSCI 49395 – Big Data Analytics and Applied Privacy, Hunter College, CUNY (Teaching Assistant) Fall 2026
- CSCI 49392 – Advanced Operating Systems, Hunter College, CUNY (Teaching Assistant; independently delivered lectures) Spring 2026
- CSCI 127 – Introduction to Computer Science, Hunter College, CUNY (Teaching Assistant) Fall 2025

Professional Service

Reviewer

- EMNLP 2026 Industry Track 2026
- ICLR 2026 Workshop on Memory in Agentic AI (MemAgents) 2026
- First Workshop on Agent Skills (AgentSkills), ACM CAIS 2026 2026
- AAAI 2025 Undergraduate Consortium (UC) 2025

Skills

• Programming	Python, Java, C/C++, JavaScript, SQL, Bash, LaTeX.
• Frameworks & Tools	LangChain, CybORG, Wireshark, Kali Linux, Git, Docker, GCP.
• Security	Network Defense, LLM-based Cyber Defense, Threat Analysis.
• AI & Productivity	Prompt Engineering, Advanced AI Coding Agents (Claude Code, Codex), Skills.
• Research Methods	Mixed-Methods Study Design, Qualitative Analysis, AI-Augmented Workflow Design.